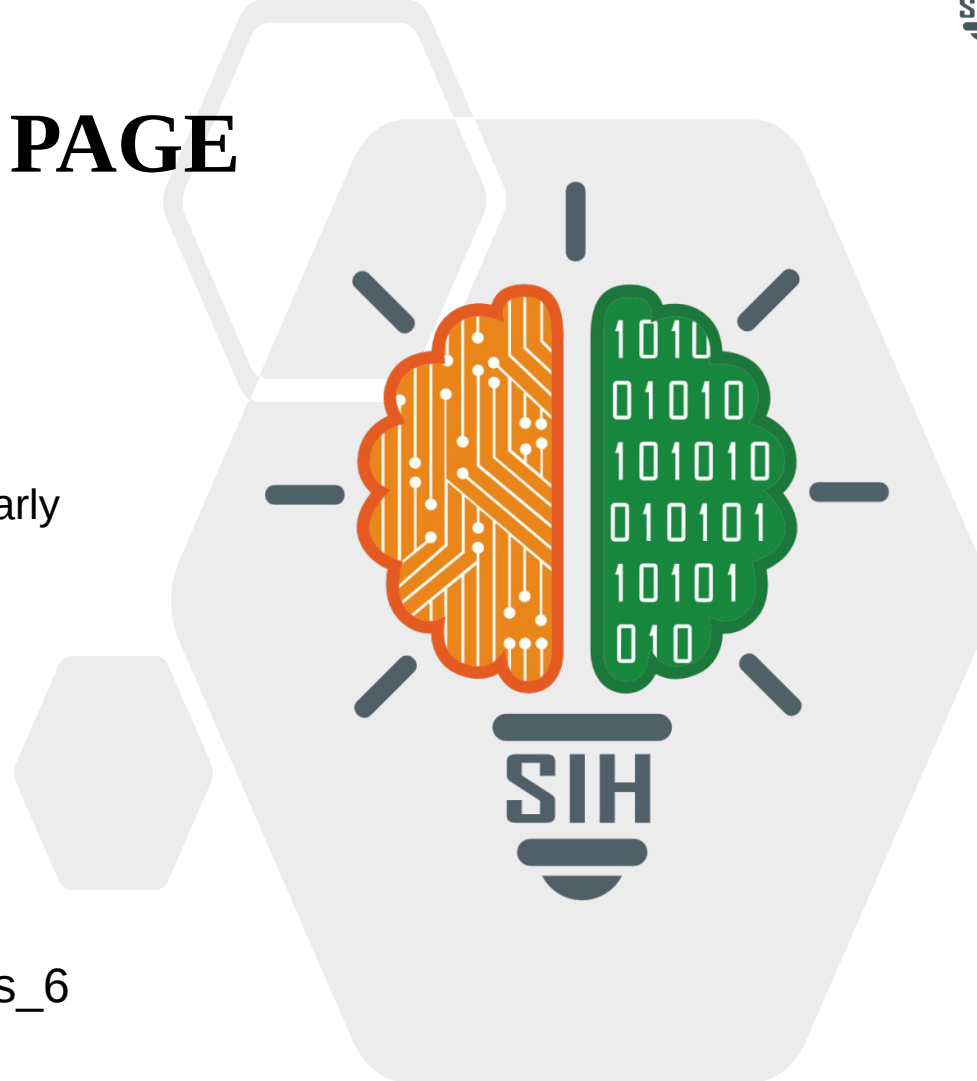




TITLE PAGE

- **Solution** – PashuRaksha
- **Problem Statement ID** – SIH26128
- **Problem Statement Title** – Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – 178601
- **Team Name (Registered on portal)** – TechDevs_6

From the first sick animal to a coordinated response, measured in hours.



Why Livestock Disease Is Found Late and Handled Slowly

1 Late reporting

Farmers try home remedies first or hide cases for fear of movement restrictions. Nobody tells them what happens next.

2 Distant diagnosis

Samples are not taken, spoil on the way, or the result never returns to the person who sent it.

3 Incomplete histories

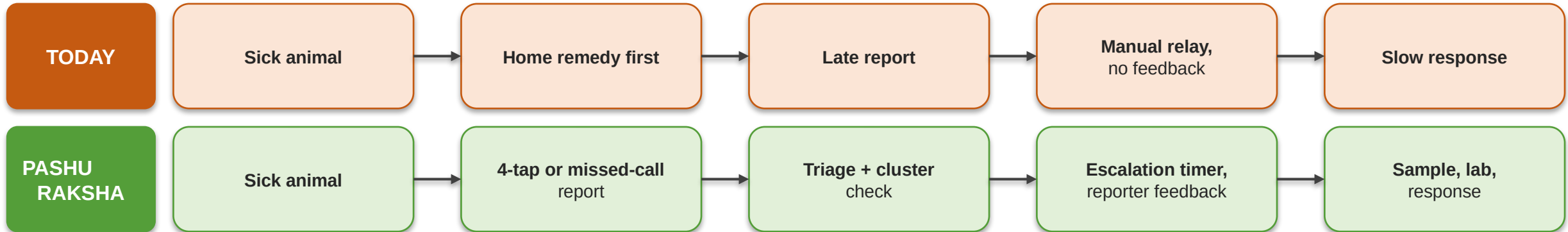
Ear-tag, vaccination and treatment records sit in paper registers, so nobody can say who is overdue.

4 Fragmented data

Dispensary, lab, drives and surveillance keep separate records; the block officer sees totals, not village patterns.

5 No signal, late zoonotic alert

Online-only tools fail where outbreaks start. Human health hears of anthrax or rabies exposure late.



The outbreak clock: hours from first report to action, recorded for every case and shown for every block.

The three jobs in the problem statement: Detect (notice the pattern early), Prevent (vaccinate and advise), Manage (sample, refer, contain, close). Most reporting apps stop at the first.

DETECT

Late reporting

Para-vet app: 4 taps, offline. Farmer: missed call, IVR in Marathi / Hindi. Assisted or anonymous reports.

Unclear severity

Syndrome rules per species give: routine advice, vet visit, or suspected notifiable disease.

Invisible patterns

Cluster thresholds across villages and weeks; baseline learned as data grows.

PREVENT

Incomplete histories

Herd record first, animal level where an ear tag exists.

Missed vaccinations

Due-lists: who in this village is overdue, with reminders.

Unseen risk

Weather and past-outbreak layer gives block-level risk guidance. Multilingual advisories.

MANAGE

Distant diagnosis

Sample checklist, nearest lab suggested, status tracked, result returned to the reporter.

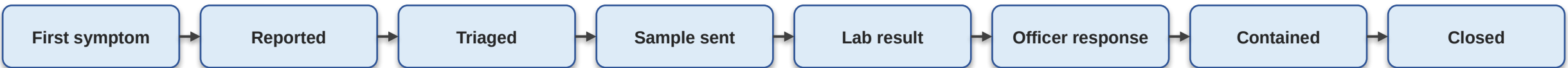
Delayed escalation

Unanswered suspect cases move up to the next officer on a timer.

Zoonotic blind spot

Zoonotic flag alerts human health. Officer dashboard shows open cases and lab turnaround.

One case, timestamped end to end (the outbreak clock)



Rules decide

Safety-critical steps: syndromes, thresholds, escalation, missing data

ML assists

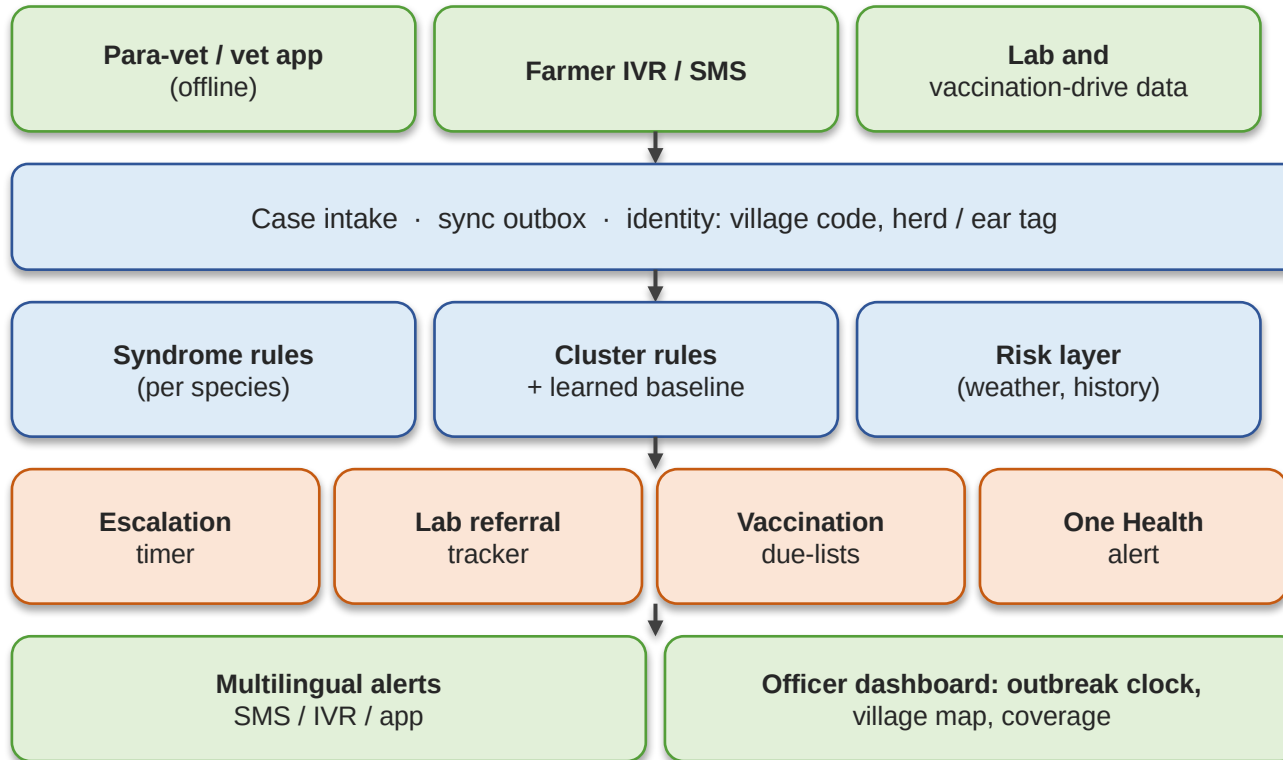
Anomaly and risk ranking where history exists; labelled as assistive

Veterinarian decides

Diagnosis, treatment and containment. The system never prescribes

Architecture and How We Will Prove It Works

Architecture



Proposed stack: Next.js PWA + Capacitor Android (offline, IndexedDB), FastAPI / Node relay, PostgreSQL + PostGIS, an IVR provider, Bhashini for language, MapLibre for the map. Nothing here has been built or benchmarked yet.

How We Will Validate

- 1 **Historical replay**
Run past outbreak reports through the rules and see whether, and when, an alert would have fired.
- 2 **Expert review**
Veterinarians check that syndrome rules and thresholds match field practice.
- 3 **Pilot in one block**
Measure the outbreak clock and the other process measures with real users.
- 4 **Holdout checks**
Test on unseen blocks and periods once enough data exists.

Proposed measures

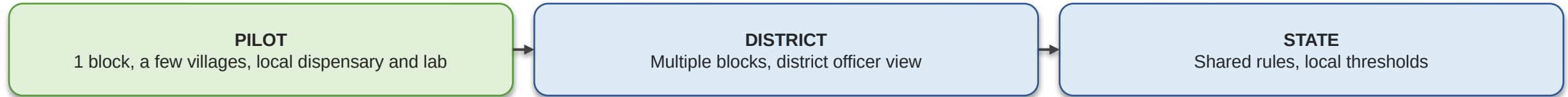
Report-to-response time

Suspects with sample sent

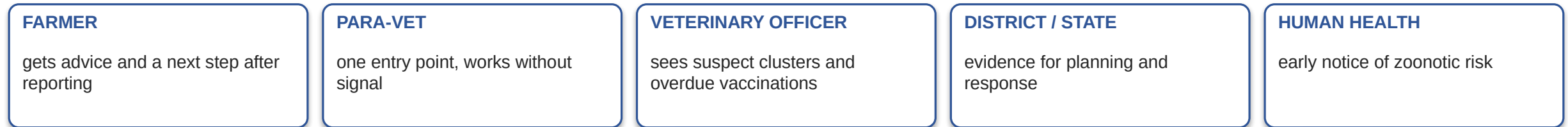
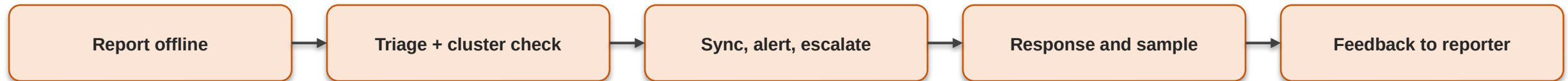
Due-list coverage

False-alarm rate

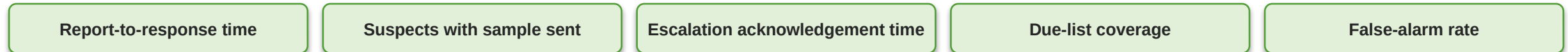
From One-Block Pilot to State-Level Surveillance



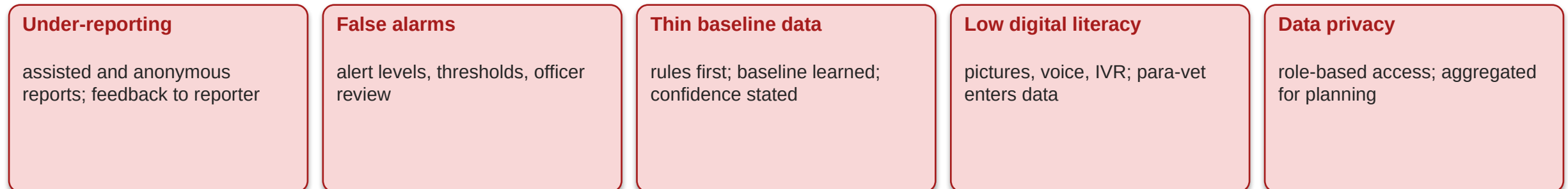
Field loop (conceptual workflow)



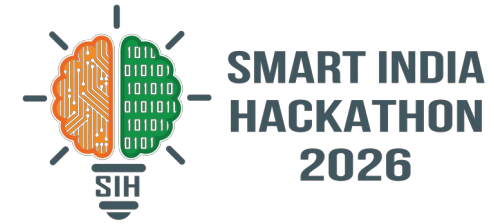
What we will measure in the pilot (proposed)



Main risks and how they are handled



RESEARCH AND REFERENCES



S. No	Reference	Published by	Used for
1	National Animal Disease Control Programme (NADCP)	Dept. of Animal Husbandry and Dairying, GoI	Vaccination-programme context
2	Livestock Census	Dept. of Animal Husbandry and Dairying, GoI	Livestock populations by area
3	Veterinary epidemiology and disease informatics	ICAR-NIVEDI	Surveillance practice, disease information
4	Bharat Pashudhan / National Digital Livestock Mission	Dept. of Animal Husbandry and Dairying, GoI	Animal ID and records integration path
5	Terrestrial Animal Health Code; notifiable diseases	WOAH	Case definitions, reporting criteria
6	One Health Joint Plan of Action	FAO, UNEP, WHO, WOAH	Zoonotic alert pathway
7	A spatial scan statistic (Kulldorff)	Communications in Statistics	Space-time cluster detection
8	Early Aberration Reporting System (Hutwagner et al.)	Journal of Urban Health	Counts-based aberration detection
9	Bhashini	MeitY	Multilingual speech and text